

**ADVANCE ROBOTICS AND AUTONOMOUS SYSTEMS FOR NEXT GENERATION
RENEWABLE AND NUCLEAR ENERGY INFRASTRUCTURE
(Applications of Robotics in Energy Generation, Inspection, Maintenance, and Safety)**

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CHAPTER ONE INTRODUCTION

1.1 Introduction to Advanced Robotics and Autonomous Systems for Energy Infrastructure

The global energy sector depends on large and complex infrastructure for the generation, conversion, transmission, and distribution of electricity. Renewable energy systems such as solar photovoltaic plants, wind turbines, hydroelectric facilities, and concentrated solar power plants operate alongside conventional and nuclear power generation infrastructure. As these facilities become more complex and geographically distributed, maintaining their reliability, safety, and operational performance requires continuous monitoring, inspection, maintenance, and fault detection.

Traditional inspection and maintenance activities frequently require personnel to work at height, in confined spaces, around high temperature equipment, underwater, or in environments containing other occupational hazards. Nuclear facilities introduce additional requirements because radiation and contamination can restrict human access to certain areas. The IAEA identifies maintenance, testing, surveillance, and inspection as essential activities for maintaining the reliability and availability of structures, systems, and components important to nuclear plant safety. ([Nucleus Apps](#))

These conditions have created increasing interest in robotics and autonomous systems for energy infrastructure. Robots can carry sensors, cameras, manipulators, and other equipment into locations that may be difficult or hazardous for human personnel to access. Research has investigated robotic applications across photovoltaic, wind, hydroelectric, concentrated solar power, power-line, and other energy infrastructure. ([DOI.org](#))

The development of advanced robotic systems therefore creates an opportunity to move energy infrastructure management from predominantly manual inspection toward **robot assisted inspection, automated monitoring, predictive maintenance, robotic intervention, and increasingly autonomous operation.**

1.2 Energy Infrastructure and the Need for Robotics

Energy infrastructure consists of interconnected physical systems used to produce and deliver energy. For electricity generation, these can include:

- solar photovoltaic plants
- wind farms
- hydroelectric plants

- concentrated solar power plants
- nuclear power plants
- electrical substations
- transmission lines
- energy-storage facilities
- supporting mechanical and electrical systems

Each type of infrastructure contains components that must remain within specified operating conditions. Example:

- **Wind energy:** Rotor → gearbox → generator → nacelle → tower → foundation
- **Solar energy:** PV modules → mounting structures → inverters → transformers → electrical distribution
- **Hydroelectric:** Water intake → turbine → generator → electrical system → transmission
- **Nuclear:** Reactor systems → heat-transfer systems → turbine → generator → electrical system

These systems require inspection and maintenance because component degradation, faults, environmental exposure, and equipment failure can affect availability and safety. This establishes the connection in this research: **Energy generation → Infrastructure → Equipment → Inspection → Maintenance → Reliability**

1.3 Renewable Energy Infrastructure

Renewable energy infrastructure uses naturally replenished energy resources such as sunlight, wind, and water. Major systems include:

1.3.1 Solar Energy

Solar photovoltaic systems convert sunlight directly into electricity through photovoltaic cells. Robotic applications can include:

- panel inspection
- thermal inspection
- surface cleaning
- vegetation monitoring
- defect detection
- autonomous site monitoring

Research on photovoltaic operation and maintenance identifies proactive maintenance and data-driven decision-making as important areas for further development. ([DOI.org](https://www.doi.org/))

1.3.2 Wind Energy

Wind turbines convert the kinetic energy of moving air into electrical energy. Major components include:

- blades
- rotor
- nacelle
- gearbox
- generator
- tower
- Foundation

Robotic systems can inspect blades, towers, nacelles, and other components. UAV based wind-turbine inspection has gained increasing attention, while other robotic systems have been investigated for internal turbine inspection. ([NREL](#))

1.3.3 Hydroelectric Energy

Hydroelectric facilities use flowing or falling water to drive turbines. Robotic systems can potentially support:

- underwater inspection
- dam inspection
- turbine inspection
- structural monitoring
- environmental monitoring

Concentrated Solar Power

Concentrated solar power systems use mirrors or other concentrating systems to focus solar radiation for thermal energy generation. Robotic systems can support:

- mirror inspection
- cleaning
- positioning
- thermal monitoring
- Maintenance

Research reviewing robotics for the energy transition has examined robotic applications across photovoltaic, wind, hydroelectric, and concentrated solar power systems. ([DOI.org](#))

1.4 Nuclear Energy Infrastructure

Nuclear power plants generate electricity through controlled nuclear reactions that produce heat, which is subsequently converted into electrical energy. The complexity of nuclear facilities creates significant requirements for:

- inspection
- surveillance
- condition monitoring
- preventive maintenance
- predictive maintenance
- corrective maintenance
- decommissioning
- radiation monitoring
- safety management

Unlike many renewable energy facilities, some nuclear environments can expose personnel and equipment to radiation and other hazardous conditions. The IAEA identifies robotics as a technology that can help access areas that are difficult or hazardous for people, including applications involving inspection, measurement, decontamination, dismantling, and handling. ([International Atomic Energy Agency](#))

This creates a second major connection: **Nuclear infrastructure → Hazardous environments → Remote operation → Robotics → Autonomous inspection and intervention**

1.5 Robotics in Energy Generation

Robotics can contribute not only to maintenance but also to the operation and monitoring of energy-generation facilities. A robotic system may perform tasks such as:

- collecting measurements
- monitoring equipment
- detecting abnormal conditions
- inspecting structures
- identifying defects
- transporting tools
- manipulating components
- cleaning surfaces
- performing repetitive maintenance activities

The robot can therefore act as a physical interface between the energy facility and the digital monitoring system. The general process becomes: **Physical Infrastructure → Sensors → Robot → Data → Analysis → Decision → Action**

1.6 Types of Robots Used in Energy Infrastructure

Different energy environments require different robotic platforms.

1.6.1 Ground Mobile Robots

They are particularly relevant for power plants and industrial facilities. These robots can move around facilities and carry:

- cameras
- thermal sensors
- radiation sensors
- LiDAR
- microphones
- environmental sensors

Recent research has examined mobile robots for nuclear power-plant inspection and security applications. ([MDPI](#))

1.6.2 Aerial Robots

They are particularly relevant for power plants and industrial facilities. UAVs can inspect:

- wind turbines
- transmission lines
- solar farms
- cooling structures
- buildings
- large industrial facilities

1.6.3 Underwater Robots

ROVs and AUVs can inspect:

- dams
- hydroelectric infrastructure
- underwater structures
- cooling-water systems
- submerged components

1.6.4 Robotic Manipulators

Manipulators can perform physical interaction with infrastructure. Applications may include:

- component handling
- tool positioning
- cleaning
- cutting
- repair
- Maintenance

Manipulators are particularly important when inspection needs to progress toward actual intervention. ([ScienceDirect](#))

1.7 Robotic Inspection

Inspection is one of the most established applications of robotics in energy infrastructure. The purpose of robotic inspection is to determine the physical or operational condition of equipment without requiring personnel to directly access every inspection location. Inspection may identify:

- cracks
- corrosion
- deformation
- overheating
- surface damage
- leaks
- material degradation
- abnormal vibration
- electrical abnormalities

A robotic inspection system can therefore replace a purely manual inspection process with a repeatable data-collection process. The general workflow becomes: **Robot Deployment** → **Sensor Acquisition** → **Data Processing** → **Defect Detection** → **Condition Assessment**

1.8 Sensors for Energy Infrastructure Robots

The effectiveness of robotic inspection depends heavily on the sensors carried by the robot. Possible sensors include:

- RGB cameras

- thermal cameras
- LiDAR
- ultrasonic sensors
- acoustic sensors
- vibration sensors
- radiation detectors;
- gas sensors
- pressure sensors
- temperature sensors
- electrical sensors

Different sensors provide different forms of information. Example:

- **Camera → Visual condition**
- **Thermal camera → Temperature distribution**
- **Ultrasonic sensor → Internal/material information**
- **Radiation sensor → Radiation level**
- **Vibration sensor → Mechanical condition**
- **LiDAR → Geometry and spatial information**

The combination of multiple sensors creates a more comprehensive representation of the infrastructure.

1.9 Computer Vision and Artificial Intelligence

Large energy facilities can generate enormous quantities of inspection data. A human operator may therefore face difficulties when manually reviewing every image, thermal scan, or sensor measurement. Artificial intelligence and computer vision can assist by identifying patterns within inspection data. Potential applications include:

- crack detection
- corrosion detection
- Blade defect detection
- solar-panel defect classification
- thermal anomaly detection
- object recognition
- structural damage detection
- radiation-map interpretation
- abnormal-condition detection

This changes the inspection process from: **Robot → Data Collection → Human Inspection** toward: **Robot → Data Collection → AI Analysis → Human Verification /**

Automated Decision. AI therefore becomes an additional layer between robotic sensing and maintenance decisions.

1.10 Autonomous Navigation

A robot operating inside an energy facility must be able to determine:

- Where am I?
- Where am I supposed to go?
- What is around me?
- How can I reach the required location safely?

Autonomous navigation can use:

- cameras
- LiDAR
- IMUs
- wheel encoders
- GPS/GNSS where available
- radar
- ultrasonic sensors
- SLAM
- sensor fusion

In nuclear facilities, autonomous navigation is being studied together with robotic inspection because direct testing inside operational plants can be difficult. Digital-twin environments have consequently been proposed for developing and evaluating robot navigation and interaction. ([Frontiers](#))

1.11 Robotic Inspection and Condition Monitoring

Inspection provides a snapshot of infrastructure conditions. Condition monitoring goes further by repeatedly collecting information over time. Example: **Inspection 1** → **Inspection 2** → **Inspection 3** → **Inspection 4**. The system can compare measurements to determine whether a component is:

- stable
- degrading
- developing an abnormal condition
- approaching a maintenance threshold

This creates the connection: **Inspection** → **Monitoring** → **Historical Data** → **Degradation Trend**

1.12 Predictive Maintenance

Predictive maintenance uses condition information to determine when equipment may require maintenance. The IAEA distinguishes predictive or condition based maintenance from periodic and corrective maintenance. Predictive maintenance involves monitoring and diagnosis to anticipate equipment failure or degradation. ([Nucleus](#))

Robotics can contribute by repeatedly collecting data from difficult to access components. Example:

- Robot Inspection
- Condition Data
- AI/Data Analysis
- Degradation Detection
- Failure Prediction
- Maintenance Planning

This is important because it moves robotics beyond simply **finding defects** toward supporting **maintenance decisions**.

1.13 Robotic Maintenance and Intervention

Inspection alone does not repair a damaged component. The next stage is robotic intervention. Robotic maintenance can involve:

- cleaning
- component manipulation
- tool operation
- surface treatment
- cutting
- fastening
- component replacement
- decontamination
- repair operations

However, intervention is considerably more complex than inspection because the robot must physically interact with the environment. This requires:

- precise positioning
- force control
- manipulation
- tool control
- environmental perception

- motion planning
- safety monitoring

Current literature on robotic energy-infrastructure maintenance identifies a gap between the large amount of research on inspection and the smaller proportion addressing actual maintenance tasks. ([ASCE Library](#))

1.14 Robotics for Nuclear Safety

Safety represents a particularly important application of robotics in nuclear infrastructure. Robots can provide information from areas where human access is restricted. Potential functions include:

- radiation measurement
- environmental monitoring
- equipment inspection
- structural inspection
- remote manipulation
- decontamination
- facility mapping
- emergency assessment

The IAEA has documented the use and development of robotic systems for nuclear decommissioning, including mobile robots equipped with sensors and three-dimensional scanning systems. ([International Atomic Energy Agency](#))

The fundamental relationship is: **Hazardous Environment → Remote Robotic Access → Data Collection → Human/AI Assessment → Safe Intervention.**

1.15 Robotics for Renewable Energy Safety

1.15.1 Wind

Robots can reduce the need for personnel to work at height during certain inspection activities.

1.15.2 Solar

Autonomous systems can inspect large solar farms while reducing repetitive manual inspection activities.

1.15.3 Hydroelectric

Robotic systems can access underwater or difficult structural locations.

1.15.4 Power Transmission

UAVs and other robotic systems can inspect transmission infrastructure without requiring personnel to physically access every structure.

1.16 Autonomous Energy Infrastructure Monitoring

The combination of robotics, sensors, AI, navigation, and communication creates the possibility of continuous or scheduled autonomous monitoring. Instead of: **Human → Periodic Inspection** the system can operate as: **Robot → Scheduled Inspection → Data Collection → AI Analysis → Condition Assessment → Alert → Maintenance Planning**.

This creates a shift from reactive maintenance toward increasingly **data driven and condition-based maintenance**.

1.17 Human Robot Collaboration

Full autonomy is not necessarily required for every energy-infrastructure task. Some operations may use **Human + Robot** rather than **Human vs Robot**. For example, a robot can enter a hazardous area and collect information while a human operator remains at a safe location and supervises the operation. The human may:

- approve actions
- review inspection results
- control the robot when necessary
- intervene during abnormal situations

This creates a **human supervised autonomous system** in which autonomy handles routine operations while human operators remain available for complex decisions.

1.18 Autonomous Multi-Robot Systems

Future energy facilities may use multiple robots rather than one robotic platform. Example:

- **UAV → External inspection**
- **Ground robot → Facility inspection**
- **Manipulator → Maintenance**
- **Underwater robot → Submerged inspection**

These robots could share information through a common energy infrastructure database or digital twin. A multi-robot system could therefore distribute inspection tasks according to:

- location
- robot capability
- available energy
- sensor requirements
- mission priority
- environmental conditions

Research into nuclear facility digital twins has already considered scenarios involving teams of robots performing inspection tasks. ([Frontiers](#))

1.19 Current Trends in Robotics for Renewable and Nuclear Energy

1.19.1 Autonomous Inspection

Robots increasingly perform inspection missions with reduced direct human control.

1.19.2 AI Based Defect Detection

Computer vision and machine learning are being applied to inspection data.

1.19.3 Predictive Maintenance

Inspection data is increasingly being used to understand equipment degradation before failure.

1.19.4 Robotic Manipulation

Research is moving from observation toward physical intervention.

1.19.5 Digital Twins

Digital representations of facilities are being used for simulation, monitoring, and robotic-system development.

1.19.6 Multi Robot Systems

Multiple robotic platforms can potentially cooperate to cover different areas and tasks.

1.19.7 Remote and Hazardous-Environment Robotics

Robotic systems are increasingly being developed for environments where human access is difficult or hazardous.

1.19.8 Data Driven Energy Infrastructure

Robotic inspection is increasingly connected to databases, AI, digital twins, and maintenance-management systems.

A recent review of energy infrastructure robotics identified inspection as more prevalent than maintenance and highlighted the lack of general-purpose robotic platforms and the continued reliance on simulation in much of the research. ([ASCE Library](#))

1.20 Future Trends in Energy Robotics

- autonomous inspection robots
- robotic maintenance systems
- AI-based predictive maintenance
- multi-robot cooperation
- autonomous UAV inspection
- advanced mobile robots
- robotic manipulators
- digital twins
- autonomous navigation
- intelligent sensor fusion
- continuous infrastructure monitoring
- adaptive robotic systems
- human-supervised autonomy

Recent work on turbine architecture and robotic maintenance is also exploring bio-inspired mobility, perception, autonomy, inspection, and in-situ repair for complex energy infrastructure. ([ScienceDirect](#))

1.21 Challenges in Robotic Energy Infrastructure

1.21.1 Harsh Operating Environments

Energy infrastructure can expose robots to:

- heat
- radiation
- water
- dust

- vibration
- electromagnetic conditions
- confined spaces
- strong winds
- difficult terrain

Nuclear robotics introduces additional requirements concerning radiation tolerance, deployment, remote maintenance, and reliability. ([MDPI](#))

1.21.2 Navigation

Robots may need to navigate complex facilities where GPS/GNSS is unavailable.

1.21.3 Perception

Sensors can be affected by:

- lighting
- dust
- water
- radiation
- reflections
- occlusion
- temperature

1.21.4 Energy

Mobile robots must carry sufficient energy to perform their missions.

1.21.5 Communication

Remote facilities may have limited or unreliable communication.

1.21.6 Manipulation

Physical maintenance requires accurate force and motion control.

1.21.7 AI Reliability

AI based inspection systems must be sufficiently reliable when used to support important maintenance or safety decisions.

1.21.8 System Integration

Robotics, AI, sensors, navigation, control, energy management, and infrastructure management systems must operate together.

1.21.9 Verification and Validation

Autonomous systems must be tested extensively before being trusted in critical infrastructure.

1.22 Research Objectives

- Investigate the applications of robotics in renewable and nuclear energy infrastructure.
- Examine robotic technologies used for inspection and monitoring of energy generation facilities.
- Investigate sensor and perception technologies for detecting defects, abnormalities, and changes in infrastructure conditions.
- Examine autonomous navigation and control techniques for robotic operation within complex energy facilities.
- Investigate the application of artificial intelligence and computer vision for automated energy infrastructure inspection.
- Examine the use of robotics and condition monitoring for predictive maintenance.
- Investigate robotic manipulation and intervention technologies for maintenance activities.
- Examine the role of robotics in improving safety within hazardous and difficult to access energy environments.
- Investigate the application of digital twins and simulation for developing and testing autonomous energy infrastructure robots.
- Examine the integration of perception, navigation, AI, inspection, maintenance, and safety within an autonomous robotic energy-infrastructure system.
- Develop or simulate a conceptual robotic system for a selected renewable or nuclear energy infrastructure application.
- Evaluate the proposed system based on inspection performance, navigation, defect detection, maintenance capability, safety, autonomy, and operational efficiency.

1.23 Research Questions

- What are the current applications of robotics in renewable and nuclear energy infrastructure?
- What robotic platforms are suitable for energy-generation, inspection, monitoring, and maintenance tasks?
- What sensing and perception technologies can be used to detect defects and abnormal conditions?

- How can autonomous robots navigate complex energy facilities?
- How can AI and computer vision improve automated energy infrastructure inspection?
- How can robotic inspection data support predictive maintenance?
- How can robotic manipulators be used for maintenance and intervention?
- How can robotics improve safety in hazardous energy environments?
- How can digital twins and simulation support the development of autonomous energy infrastructure robots?
- How can perception, navigation, AI, inspection, maintenance, and safety be integrated into a unified robotic system?
- What challenges limit the deployment of autonomous robotic systems in renewable and nuclear energy infrastructure?

1.24 Significance of the Research

The significance of this research lies in examining how robotics can become integrated into the operation and maintenance lifecycle of energy infrastructure.

Rather than considering robotics only as a machine for performing an individual inspection task, the research considers robotics as part of a larger system connecting: **Energy Generation → Monitoring → Inspection → Intelligence → Maintenance → Safety**

This provides a framework for investigating how autonomous robotic systems can interact with complex energy infrastructure while collecting information, identifying abnormal conditions, supporting maintenance decisions, and performing selected physical tasks.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Renewable and nuclear energy facilities contain large and complex infrastructure that requires regular inspection, monitoring, maintenance, and safety assessment. Examples include wind turbines, photovoltaic plants, hydropower facilities, nuclear power plants, and associated equipment. Conventional inspection and maintenance methods can require workers to operate at height, in confined spaces, in difficult terrain, or in environments where exposure to radiation and other hazards must be controlled. Robotics has therefore attracted increasing attention as a means of performing inspection and maintenance tasks while reducing direct human exposure and improving the frequency and consistency of infrastructure monitoring [1], [2].

Recent research has demonstrated the use of aerial, ground, climbing, underwater, and robotic manipulation systems across different energy applications. These systems combine sensors, perception, navigation, control, artificial intelligence, and communication technologies to observe infrastructure and identify abnormal conditions. However, much of the existing work remains focused on individual tasks such as inspection, defect detection, or navigation rather than the complete integration of these functions into one autonomous system [1]. This chapter therefore reviews robotic applications in renewable and nuclear energy infrastructure, with emphasis on inspection, sensing, autonomous navigation, artificial intelligence, predictive maintenance, manipulation, safety, digital twins, system integration, and performance evaluation.

2.2 Applications of Robotics in Renewable and Nuclear Energy Infrastructure

Robotics is being applied across renewable and nuclear energy facilities because many infrastructure components are large, distributed, difficult to access, or potentially hazardous to inspect manually. In renewable energy, robotic systems have been investigated for wind turbine blade inspection, photovoltaic plant monitoring, and other maintenance activities. For example, UAVs equipped with cameras and thermal sensors can inspect large photovoltaic installations and identify abnormal modules without requiring personnel to physically examine every panel [3]. Ground robots can also operate closer to PV modules and inspect areas that may be difficult for aerial platforms to observe.

Wind energy has similarly benefited from robotic inspection. UAVs and other robotic platforms can capture high-resolution images of turbine blades, allowing defects such as cracks, erosion, and coating damage to be identified using image-processing and deep-learning techniques [4]. Ground and climbing robots provide additional options for inspecting components that are difficult to reach with conventional methods.

Nuclear facilities present a different operating environment because inspection and maintenance may involve radiation, restricted access, contaminated areas, and complex plant structures. Research has therefore investigated mobile robots, remotely operated systems, and robotic manipulators for inspection, emergency response, decommissioning, and maintenance [5], [6]. The development of mobile robots for nuclear power plant inspection has also included autonomous mapping and navigation systems capable of operating within plant environments [7].

These applications show that robotics can support different stages of energy infrastructure operation, from routine inspection to maintenance and hazardous-environment intervention. However, the requirements of renewable and nuclear facilities differ considerably, meaning that robotic systems must be selected according to the infrastructure, environment, inspection task, and level of autonomy required.

2.3 Robotic Technologies for Inspection and Monitoring

Inspection is one of the most established applications of robotics in energy infrastructure. Robotic platforms can carry cameras, thermal sensors, LiDAR, ultrasonic sensors, radiation detectors, and other instruments to collect information about infrastructure without requiring continuous human access.

For photovoltaic plants, UAV-based inspection has been investigated using visible and thermal imaging. Thermal images can reveal abnormal temperature patterns associated with faulty or underperforming PV modules, while visual images can identify physical damage and surface abnormalities [3], [8]. Ground robots can complement UAV inspection by accessing areas beneath or between PV arrays that may be difficult to observe from the air [9].

Wind turbine inspection similarly uses robotic platforms equipped with high-resolution imaging systems. UAVs can inspect turbine blades from different viewing angles, while climbing and other robotic systems can provide closer inspection of blade surfaces and structural components [4]. These approaches can reduce the need for manual inspection at height and allow inspection data to be collected more frequently.

In nuclear facilities, mobile robots can be used to inspect plant areas while carrying cameras and other sensors. Będkowski demonstrated an end-to-end navigation system for nuclear power plant inspection using a mobile robot equipped with LiDAR and inertial sensing for mapping and localization [7]. Such systems demonstrate how inspection robotics is increasingly moving beyond simple remote viewing toward autonomous data collection.

2.4 Sensor and Perception Technologies for Energy Infrastructure

The effectiveness of an inspection robot depends heavily on its ability to perceive both its surroundings and the condition of the infrastructure being inspected. Different sensors provide different forms of information. RGB cameras provide visual information for identifying surface defects, while thermal cameras can detect temperature abnormalities. LiDAR provides three-dimensional information useful for mapping and obstacle detection, while ultrasonic and other non-destructive testing technologies can provide information about internal or subsurface conditions.

In wind turbine inspection, high-resolution cameras have been widely investigated for detecting blade cracks, erosion, and surface damage, while deep-learning models can process these images automatically [4]. PV inspection similarly combines visible and thermal imagery to identify physical and electrical abnormalities [3], [8].

Nuclear facilities require additional sensing capabilities because the robot may need to detect environmental and radiological conditions in addition to physical infrastructure defects. Research on nuclear mobile robots therefore considers perception, protection, mapping, and planning as interconnected functions [6].

Sensor fusion can improve the reliability of perception by combining information from multiple sensors rather than depending on one measurement source. For example, a robot may combine camera information with LiDAR for localization and obstacle detection while using thermal or ultrasonic sensing for infrastructure assessment. The selection and combination of sensors therefore depend on the inspection objective, operating environment, and type of defect being investigated.

2.5 Autonomous Navigation and Control in Energy Facilities

Energy facilities can contain narrow passages, equipment, structural obstacles, uneven surfaces, restricted areas, and changing environmental conditions. A robotic inspection system must therefore be capable of determining its position, planning movement, avoiding obstacles, and maintaining stable operation.

Autonomous navigation commonly combines localization, mapping, path planning, and motion control. SLAM techniques allow a robot to construct a map while estimating its position within the environment. Będkowski developed a navigation system for nuclear power plant inspection using LiDAR and inertial measurement information to support simultaneous mapping and localization [7]. The system demonstrates the application of real-time localization and mapping to the inspection of complex nuclear environments.

Navigation requirements vary according to the robotic platform. UAVs may require three-dimensional navigation around turbine blades and solar installations, while ground robots require reliable localization and obstacle avoidance within plant structures. In nuclear facilities,

navigation must also account for restricted or hazardous areas where reliable autonomous operation is particularly important.

Control systems provide the connection between the navigation system and the physical movement of the robot. The controller must translate planned movements into appropriate motor or actuator commands while maintaining stability and avoiding collisions. Consequently, navigation and control cannot be treated independently when designing an autonomous energy infrastructure robot.

2.6 Artificial Intelligence and Computer Vision for Automated Inspection

Artificial intelligence and computer vision have become important tools for converting inspection data into useful information. Instead of requiring an operator to examine every image manually, machine-learning and deep-learning methods can identify patterns associated with defects or abnormal conditions.

In wind energy, deep learning approaches have been investigated for detecting blade cracks, erosion, and other surface defects from UAV imagery [4]. Similarly, UAV-based photovoltaic inspection combines imaging techniques and data-driven analysis to identify and classify different PV failure conditions [3]. Research on PV module defect detection has also examined the integration of UAVs and intelligent diagnostic techniques for large-scale photovoltaic plants [8].

AI is also being investigated for nuclear infrastructure condition monitoring. Machine-learning methods have been applied to anomaly detection, fault diagnosis, and equipment condition assessment in nuclear power plants [10]. More recent work has combined AI and digital twins for nuclear condition monitoring, including anomaly detection, fault classification, and virtual sensing [11].

Despite these developments, AI based inspection depends heavily on the quality and representativeness of the available data. Changes in lighting, weather, viewing angle, sensor quality, environmental conditions, and defect appearance can affect detection performance. Therefore, AI should be considered as part of a larger perception and inspection system rather than as an isolated solution.

2.7 Robotics and Condition Monitoring for Predictive Maintenance

Inspection provides information about the present condition of an energy asset, while predictive maintenance attempts to use condition information to determine whether maintenance may be required in the future. This approach can reduce dependence on fixed maintenance schedules and allow intervention to be planned according to the actual condition of equipment.

In renewable energy systems, AI and machine learning are increasingly being applied to condition monitoring, anomaly detection, fault diagnosis, and remaining useful life prediction [12]. Sensor data from equipment can be analysed to identify changes that may indicate degradation before a major failure occurs.

Similar approaches are being investigated in nuclear power plants. Machine-learning-based condition assessment has been applied to anomaly detection, fault detection and diagnosis, and remaining useful life prediction for nuclear equipment [10]. Digital twins are also being investigated as a means of combining physical models, sensor data, and AI-based analytics for continuous condition monitoring [11].

Robotics can provide an additional layer to this process. A robotic system can physically collect inspection data from an asset, while AI processes the data and identifies possible abnormalities. The resulting information can then support maintenance planning. This creates a connection between **robotic inspection, condition monitoring, defect detection, and predictive maintenance**.

2.8 Robotic Manipulation and Intervention for Maintenance

Inspection alone does not address every maintenance requirement. In some applications, a robotic system must be capable of physically interacting with infrastructure. Robotic manipulation can include grasping, component handling, tool operation, cleaning, cutting, or other maintenance-related activities.

This capability is particularly important in nuclear environments where direct human intervention can be hazardous. Research into nuclear decommissioning has investigated robotic manipulators and tele-operated systems for performing complex handling operations in contaminated environments [13]. However, existing nuclear robotic manipulation has historically relied heavily on human teleoperation, with relatively limited autonomous manipulation capabilities [13].

More advanced systems can combine robotic manipulators with cameras, force sensing, motion planning, and computer vision. This can allow a robot to determine the location of an object, position its manipulator, and perform a predefined intervention. For energy infrastructure, such capabilities could eventually support tasks such as component handling, surface cleaning, inspection tool positioning, and minor maintenance operations.

The major challenge is that manipulation in energy facilities requires both accurate perception and precise control. The robot must understand the location and condition of the target before attempting an intervention. This makes robotic manipulation closely connected to the perception, navigation, and AI components of the overall system.

2.9 Robotics for Safety in Hazardous Energy Environments

Safety is one of the strongest reasons for applying robotics to energy infrastructure. Renewable energy facilities can contain elevated structures, electrical equipment, rotating machinery, remote installations, and difficult-to-access components. Nuclear facilities add further challenges associated with radiation, contamination, and restricted access.

Robots can reduce the amount of direct human exposure by allowing inspection or intervention to be performed remotely. The use of robots during nuclear accidents, including the Fukushima Daiichi response, demonstrated the importance of robotic systems for accessing environments that are difficult or dangerous for humans [5].

Safety, however, cannot be achieved simply by removing the operator from the physical environment. Autonomous robots must also operate reliably and avoid creating additional hazards. Navigation errors, sensor failures, communication loss, incorrect defect classification, or unexpected obstacles can affect the safety of the facility and the robot itself.

For this reason, safety should be incorporated into the robotic architecture through obstacle detection, collision avoidance, fault handling, controlled motion, emergency behaviours, and appropriate levels of human supervision.

2.10 Digital Twins and Simulation for Robotic Energy Infrastructure

Physical testing of robotic systems inside operational energy facilities can be expensive, difficult, and potentially unsafe. Simulation and digital twins therefore provide an important environment for developing and testing robotic systems before physical deployment.

A digital twin provides a virtual representation of a physical system that can be connected to operational data. Recent research has examined digital twins across renewable energy, nuclear energy, and other clean energy systems for monitoring, simulation, control, maintenance, and decision-making [14]. Digital twins can also provide virtual environments in which robotic systems can be tested under different operating conditions.

In nuclear applications, digital twins are being developed specifically for robotics and autonomous intelligence. The iFANpp system, for example, provides a full-scale nuclear power plant digital twin intended to support robotic simulation, AI research, reinforcement learning, and teleoperation [15].

Simulation therefore provides a practical route for testing navigation, perception, planning, control, defect detection, and maintenance scenarios without immediately requiring access to an operational facility. This is particularly relevant to the proposed study because the conceptual robotic system can first be developed and evaluated within a simulated energy infrastructure environment.

2.11 Integration of Perception, Navigation, AI, Inspection, Maintenance and Safety

The literature shows that perception, navigation, AI, inspection, maintenance, and safety are often investigated as separate research areas. However, an autonomous energy infrastructure robot requires these functions to operate together.

A typical integrated process begins with sensors collecting information about the robot's surroundings and the infrastructure. Perception algorithms process this information to identify objects, obstacles, and possible defects. Localization determines the robot's position, while the navigation system generates a suitable route to the inspection location. The robot then moves under the control system while continuously updating its perception of the environment.

Once inspection data are collected, computer vision or other AI methods can identify abnormalities. The condition information can then be used to determine whether further inspection or maintenance is required. If physical intervention is possible, the robot can use its manipulation system to perform the required task. Safety monitoring operates throughout the process to detect obstacles, abnormal conditions, or system failures.

This can be represented as: **Energy Infrastructure → Sensors → Perception → Localization → Navigation → Path Planning → Control → Inspection → AI/Defect Detection → Condition Assessment → Maintenance Decision → Robotic Intervention** with **Safety Monitoring** and **Human Supervision** operating across the system.

Recent reviews of robotic energy infrastructure identify a similar challenge: inspection applications are considerably more developed than general-purpose autonomous maintenance systems, while many proposed systems remain dependent on simulation rather than extensive real-world deployment [1]. This provides a basis for investigating the integration of these functions within a single conceptual architecture.

2.12 Conceptual Robotic System for Energy Infrastructure

Based on the reviewed literature, a conceptual autonomous robotic system can be developed for a selected renewable or nuclear energy application. The specific platform may be selected according to the chosen infrastructure, such as a mobile robot for nuclear facility inspection, a UAV for wind turbine inspection, or a ground robot for photovoltaic plant monitoring.

The system would contain a robotic platform, sensing subsystem, onboard computing, localization and navigation modules, inspection sensors, AI-based defect detection, control system, communication interface, and safety monitoring functions. A digital-twin or simulation environment can be used to represent the selected energy facility and provide a controlled environment for testing.

The operational sequence would begin with a defined inspection mission. The robot would localize itself within the environment, perceive surrounding infrastructure, generate a path to the inspection area, and execute the movement using the control system. Inspection sensors would then collect infrastructure data, while AI-based algorithms would analyse the data for defects or abnormalities. If a maintenance requirement is identified, the system could either generate a maintenance recommendation or, where the selected robotic platform supports it, perform a defined intervention. This conceptual system provides the foundation for the methodology and simulation phase of the research.

2.13 Performance Evaluation of Autonomous Energy Infrastructure Robots

The performance of an autonomous robotic energy infrastructure system should be evaluated using measurable technical criteria rather than only determining whether the robot completes a task.

Inspection performance can be evaluated using inspection coverage, inspection time, sensor data quality, and repeatability. **Navigation performance** can be measured using localization error, trajectory error, navigation success, path length, and obstacle-avoidance performance.

For AI-based inspection, appropriate measures include defect-detection accuracy, precision, recall, F1-score, and false-positive and false-negative rates. Maintenance capability can be evaluated based on task completion, manipulation accuracy, intervention time, and successful execution of the selected maintenance task.

Safety can be evaluated through collision avoidance, emergency response, obstacle detection, and operation under simulated hazardous conditions. Overall autonomy can be assessed by measuring how much of the mission can be completed without direct operator intervention. Operational efficiency can include mission duration, computational requirements, energy consumption, and successful completion of the assigned inspection mission.

These metrics provide a clear connection between the literature review and the experimental methodology because each major research objective can be translated into a measurable performance criterion.

2.14 Research Gap

The literature demonstrates significant progress in the use of robotics for renewable and nuclear energy infrastructure. UAVs, ground robots, climbing robots, and robotic manipulators have been investigated for inspection, monitoring, and maintenance. Computer vision and AI have improved automated defect detection, while SLAM, LiDAR, inertial sensing, and other technologies have supported autonomous navigation. Digital twins have also created new

opportunities for developing and testing robotic systems before physical deployment [1], [7], [11], [15].

However, the literature remains fragmented across individual applications and technologies. A recent review of robotic inspection and maintenance of energy infrastructure identified the continued dominance of inspection over maintenance, the lack of general-purpose robotic platforms, and the reliance on simulation compared with real-world deployment [1]. Similarly, nuclear robotic research continues to identify challenges associated with perception, planning, protection, manipulation, and autonomous operation [6], [13].

Therefore, a research gap exists in the **integration of these technologies into a complete autonomous robotic system for energy infrastructure**. Perception must provide information for navigation and inspection; navigation must enable the robot to reach inspection locations; AI must interpret collected data; condition monitoring must support maintenance decisions; manipulation must enable intervention where required; and safety must operate throughout the entire mission.

The proposed research addresses this gap by developing or simulating a conceptual robotic system for a selected renewable or nuclear energy infrastructure application and evaluating its inspection, navigation, defect detection, maintenance, safety, autonomy, and operational performance.

2.15 Summary

This chapter reviewed the application of robotics in renewable and nuclear energy infrastructure, beginning with robotic inspection and monitoring and progressing through sensing, perception, navigation, control, AI-based defect detection, predictive maintenance, robotic manipulation, safety, and digital twins. The literature shows that individual technologies have reached different levels of development, with inspection and monitoring currently receiving substantial attention.

The review also shows that the major challenge is no longer simply developing individual robotic technologies but determining how these technologies can operate together within an autonomous energy infrastructure system. The interaction between perception, navigation, inspection, AI, maintenance, manipulation, and safety provides the basis for the proposed system architecture. The identified research gap therefore leads directly into the methodology, where a selected energy infrastructure can be modelled, a conceptual robotic system developed or simulated, and its performance evaluated using defined technical metrics.

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CHAPTER THREE

METHODOLOGY